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JEL classification: R10, R11, R58

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Abstract

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Keywords: a, b, c

1 Introduction

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Regional inequality is a pervasive feature of the Indonesian economy (Esmara 1975; Mishra 2009; Bendesa et al. 2016). To a large extent, the insular geography and the unbalanced spatial distribution of natural resources suggest that regional inequality is an expected outcome. However, regional improvements in labor productivity may help reduce these regional imbalances and promote economic development. Moreover, since the early 2000s, major political reforms such as decentralization and democratization initiatives may have influenced the trajectories of labor productivity and its proximate determinants: physical capital, human capital, and aggregate efficiency.

The results of this paper contribute to the literature of regional development in Indonesia in three fronts. First, there is a large literature that studies provincial income disparities in Indonesia (Akita 1988; Mishra 2009; Akita, Kurniawan & Miyata 2011). To this literature, this paper contributes an evaluation of provincial disparities in some of the main determinants of per-capita income: labor productivity, physical capital, human capital and efficiency.[1] Second, there is a growing literature that studies provincial convergence with a focus on the dynamics of the "average" province that converges to a unique equilibrium (Garcia & Soelistianingsih 1998; Resosudarmo & Vidyattama 2006; Vidyattama 2013). To this literature, this paper provides an alternative perspective the goes beyond the dynamics of the average province. Specifically, this paper incorporates the role of provincial heterogeneity and the formation of multiple convergence clubs (multiple equilibria). Third, there is an emerging literature that studies

[1] By studying the determinants of income, this paper is also related to the literature that decomposes income differences into factors in Indonesia (Akita & Lukman 1995; Kataoka 2010, 2018).

provincial convergence in Indonesia from a convergence clubs perspective (Gunawan, Mendez & Santos-Marquez 2019; Kurniawan, Groot & Mulder 2019; Mendez 2020). To this literature, this paper contributes a more comprehensive evaluation of productivity-related indicators.[2]

The rest of this paper is organized as follows. Section 2 provides an overview of the related literature. Section 3 describes the data and documents a set of stylized facts. Section 4 presents the methodological approach of the paper. Section 5 discusses the results. Finally, Section 6 offers some concluding remarks.

2 Related literature

2.1 Two established literatures: Productivity accounting and convergence

At least since the seminal contribution of Solow (1956), the study economic growth has emphasized the role of capital accumulation and aggregate efficiency (technological progress) for understanding labor productivity differences across countries and over time.

2.2 Other subsection

3 Data and stylized facts

3.1 Data construction and descriptive statistics

Data of 26 provinces over the 1990-2010 period are used to measure labor productivity and its proximate sources: capital accumulation and aggregate efficiency Based on publicly available information from the Central Bureau of Statistics of Indonesia, labor productivity is constructed by dividing real Gross Regional Domestic Product (GRDP) by the number of workers in the labor force. Based on the same data source, human capital per worker is constructed as a weighted average of the years of education of the labor force. Based on the physical capital stock estimates of Kataoka (2013), physical capital per worker is constructed by dividing provincial capital stock by the number of workers in the labor force.

[2]The papers of Sakamoto (2007) and Gunawan, Mendez & Santos-Marquez (2019), for instance, only focus on income, while the paper of Mendez (2019) only focuses on efficiency. Although Kurniawan, Groot & Mulder (2019) evaluate four different socio-economic indicators, they do not include the dynamics of labor productivity, physical capital, and efficiency, which are the main indicators of the current paper.

3.2 Regression table in plain text

4 Regression Results

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                                Dependent variable:
                    -------------------------------------------------------
                                   rating                     high.rating
                                    OLS                        probit
                          (1)                    (2)              (3)
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complaints	0.692***	0.682***	
	(0.149)	(0.129)	
privileges	-0.104	-0.103	
	(0.135)	(0.129)	
learning	0.249	0.238*	0.164***
	(0.160)	(0.139)	(0.053)
raises	-0.033		
	(0.202)		
critical	0.015	-0.001	
	(0.147)	(0.044)	
advance	-0.062		
	(0.042)		
Constant	11.000	11.300	-7.480**
	(11.700)	(7.320)	(3.570)
Observations	30	30	30
R2	0.715	0.715	
Adjusted R2	0.656	0.682	
Log Likelihood		-9.090	
Akaike Inf. Crit.		26.200	
Residual Std. Error	7.140 (df = 24)	6.860 (df = 26)	
F Statistic	12.100*** (df = 5; 24)	21.700*** (df = 3; 26)	

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Note: *p<0.1;* ***p<0.05;*** p<0.01

5 Methodology

5.1 Convergence framework

Phillips & Sul (2007) proposed a convergence test based on the decomposition of panel data. Consider a variable, y_{it}, that can be decomposed as follows:

$$y_{it} = g_{it} + a_{it}, \tag{1}$$

where g_{it} is a systematic component and a_{it} is a transitory component. To further separate common from idiosyncratic components, Equation 1 is restated as follows:

$$y_{it} = \left(\frac{g_{it} + a_{it}}{\mu_t} \right) \mu_t = \delta_{it} \mu_t, \tag{2}$$

where δ_{it} is an idiosyncratic component and μ_t is a common component.

Intuitively, δ_{it} describes the transition path of each economy towards its own equilibrium growth path and μ_t describes a hypothesized equilibrium growth path that is common to all economies. More formally, Equation 2 is a dynamic factor model where the idiosyncratic component, δ_{it}, is a factor-loading coefficient that represents the individual distance between a common trending behavior, μ_t, and the observed variable, y_{it}.

Next, the following semi-parametric specification is suggested by Phillips & Sul (2007) and Phillips & Sul (2009) to characterize the dynamics of the idiosyncratic component, δ_{it}:

$$\delta_{it} = \delta_i + \frac{\sigma_i \xi_{it}}{log\,(t)\, t^{\alpha}}, \tag{3}$$

where δ_i is constant over time but varies across economies, ξ_{it} is a weakly time dependent process with mean 0 and variance 1 across economies.

Specifically, a one-sided t test with heteroskedasticity-autocorrelation consistent (HAC) standard errors is used. In this setting, the null hypothesis of convergence is rejected when $t_b < -1.65$.

5.2 Other subsection

6 Results

The log t test of convergence suggested by Phillips & Sul (2007) rejects the convergence hypothesis for labor productivity.

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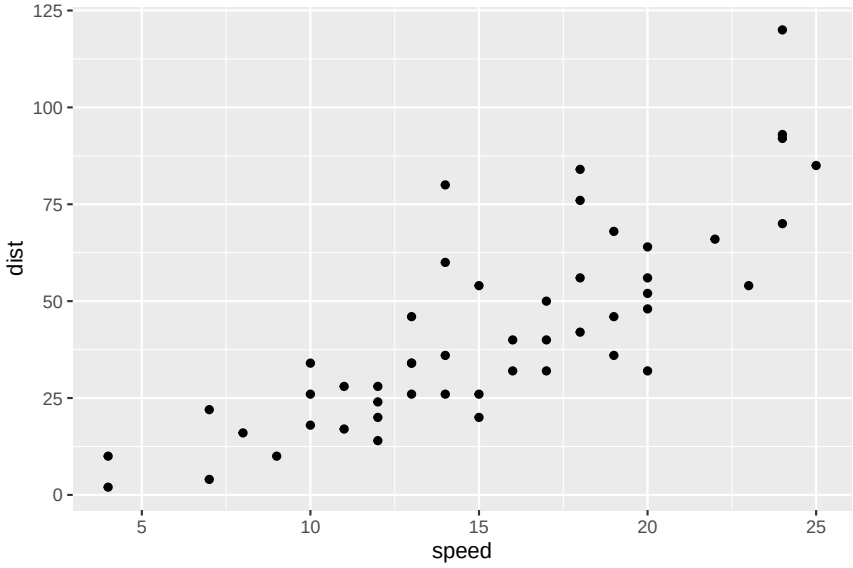

Figure 1: *Title of the figure*

Table 2: *A knitr kable table*

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4	2
4	10
7	4
7	22
8	16

Table 3: *An imported table*

X1	Mean	Std. Dev.	Min	Median	Max
Scale 1: Original values	NA	NA	NA	NA	NA
Labor productivity	19.09	16.82	5.89	14.35	75.03
Physical capital	42.39	48.67	11.13	27.27	248.79
Human capital	7.91	0.99	5.27	7.95	10.42
Efficiency (parametric)	1.33	0.57	0.73	1.21	3.35
Efficiency (nonparametric)	77.13	17.76	40.70	77.55	100.00
Scale 2: Relative log trends	NA	NA	NA	NA	NA
Labor productivity	1.00	0.22	0.64	0.96	1.59
Physical capital	1.00	0.21	0.70	0.96	1.61
Human capital	1.00	0.06	0.82	1.01	1.14
Efficiency (parametric)	1.00	1.52	-1.33	0.82	5.38
Efficiency (nonparametric)	1.00	0.06	0.86	1.01	1.07

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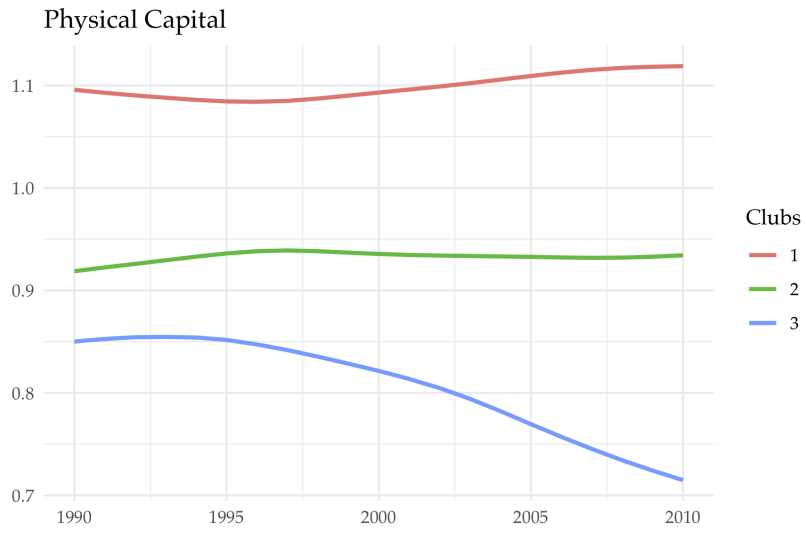

Figure 2: *Figure from png*

7 Concluding remarks

This paper studies the evolution of provincial disparities in labor productivity, physical and human capital accumulation per worker, and aggregate efficiency in Indonesia over the 1990-2010 period. In particular, the convergence test proposed by Phillips & Sul (2007) is applied to evaluate whether all provinces are converging to a common steady-state path. The results are three fold. First, there is a lack of overall convergence in labor productivity and two convergence clubs characterize its provincial dynamics. Second, the hypothesis of overall convergence is also rejected for both capital inputs. Physical and human capital per worker are characterized by three and two convergence clubs, respectively. Third, aggregate efficiency is the only production variable for which the convergence hypothesis is not rejected.

Appendix

A) Convergence clubs using a common Y axis

Labor productivity

Physical capital per worker

B) Initial (before merge) convergence clubs of physical capital per worker

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